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SUPERSEDING
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FEDERAL SPECIFICATION

PAPER, STENCIL, DUPLICATING-MACHINE, DRY-PROCESS

This specification was approved by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government.

1. CLASSIFICATION

1.1 Type. — Stencil paper covered by this specification shall be of but one type.

1.2 Sizes and scale numbers. — Unless otherwise specified, the stencil paper shall be of the following sizes and scale numbers, as specified:

8½ x 14; scale numbers 1 to 62, inclusive.
8½ x 18; scale numbers 1 to 84, inclusive.
A tolerance of plus or minus 1/16 inch in each direction of the paper will be permitted.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 The following Federal Specifications, of the issues in effect on date of invitation for bids, form a part of this specification:

TT-C-621—Correction-Compound; Stencil.
UU-P-31 — Paper; General Specifications and Methods of Testing.
UU-P-388—Paper; Mimeograph.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.—The stencil paper shall be of suitable fibrous tissue coated with non-wax-base stencil-forming material.

3.2 Odor.—The stencil paper shall be free from any pronounced disagreeable odor.

3.3 Transparency.—The stencil paper shall have sufficient transparency to permit satisfactory visibility of guides placed under it.

3.4 Tensile breaking strength.—The tensile breaking strength, average of both directions of the paper, shall be not less than 2.4 pounds per inch width.

3.5 Elongation.—The elongation, average of both directions of the paper, shall be not more than 6 percent.

3.6 Moisture resistance.—The paper shall withstand the test specified in 4.3.3.

3.7 Permanence. — The paper shall withstand the tests specified in 4.3.4 and 4.3.5 and shall retain its serviceability, as defined by this specification, under normal storage conditions for not less than 12 months from time of delivery.

3.8 Cut-outs and type filling.—The paper shall withstand the tests specified in 4.3.6 and 4.3.7.

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3.9 Correction properties.—There shall be no noticeable difference in the quality of duplication by stencils before and after correction compound has been applied as specified in 4.3.8.

3.10 Use with stylus.—Stencils shall show complete and uniform displacement without tearing or pulling the paper, and without requiring a protecting sheet on top of the stencil, when tested with a stylus as specified in 4.3.8.

3.11 Duplicating quality. — Stencils made from the stencil paper shall be capable of producing not less than 5,000 clear copies without distortion or other failure that impairs legibility. The cuts in the stencils shall be readily penetrated by ink to give clear distinct copies. There shall be no noticeable difference in depth of print between the 25th and 100th copy, between the 100th and 500th copy, between the 500th and 1000th copy, and each additional 500th copy up to and including the 5000th copy. (See 4.3.8.) When specified, the quality of the 5000th print shall be equal in all respects to a comparison print accompanying the invitation for bids.

3.12 Legibility of typing.—The finish of the stencil paper shall be such as to permit easy reading of stencils made from it. (See 4.3.9.)

3.13 Backing sheet.—Each sheet of stencil paper shall be properly fastened to a suitable backing sheet. The backing sheet shall be perforated or indented at suitable distances near the upper end to permit easy separation of the body of the backing sheet from the stub which holds the stencil sheet in the duplicating machine. The stencil paper shall be so fastened to the stub that it cannot be removed without being torn. The top of the backing sheet shall contain four holes suitable for holding the stub in the duplicating machine.

3.14 Cushion sheets. — Not less than 8 sheets of coated or impregnated cushion sheets or 12 sheets of noncoated or nonimpregnated cushion sheets per 24 sheets of stencil paper shall be furnished. The cushion sheets shall adequately hold and remove the stencil-forming material displaced during the cutting of the stencil.

3.15 Workmanship. — The stencil paper shall comply with a comparison sample furnished with the invitation for bids with respect to finish, formation, shives, holes, and uniformity of coating.

3.16 Instruction sheets.—Detailed instructions, giving the proper use of the stencil sheets, shall be furnished with each package of stencils. (See 5.1.)

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—The sample for inspection and testing shall consist of at least 3 packages taken from each lot of 1,000 packages or fraction thereof delivered, except that the total sample shall not exceed 15 packages.

4.2 Inspection.—Samples shall be inspected for size, scale numbers, material, odor, transparency, backing sheet, cushion sheets, workmanship, packaging, packing, and marking. (See 1.2, 3.1, 3.2, 3.3, 3.13, 3.14, 3.15, 5.1, 5.2, and 5.3.)

4.3 Tests.

4.3.1 The following tests shall be made in accordance with methods described in Federal Specification UU–P–31:

Test	*Method*
Atmospheric conditions	102
Tensile breaking strength (dry) and elongation	171

4.3.2 *Mimeograph paper and cushion sheet.* —The paper used for all mimeographing tests shall comply with the requirements for type III, 40-pound chemical wood paper specified in Federal Specification UU–P–338. The cushion sheets furnished with the stencil paper shall be used in making the stencils.

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4.3.3 *Moisture resistance.*—Place a drop of distilled water at room temperature (about 25° C.) on the backing sheet of the stencil paper (between the stencil paper and backing sheet). Press the coated side of the stencil paper in contact with the drop of distilled water, using gentle pressure. Allow to remain in contact for 5 minutes and then separate with a steady pull. The stencil paper shall be easily separated from the backing sheet without any indication of sticking or disintegration.

4.3.4 *Accelerated aging.*—Take 20 alternate strips each of the stencil paper and backing, ½ inch x 4 inches, and place them between a pair of wooden blocks 1 inch x 4 inches x ½ inch which have been previously dried at 105° ± 2° C. Place each assembly in an oven in which the temperature is maintained uniform throughout, with a 2-pound weight on each pair of blocks, and heat for 3 hours at 105° C. (± 2° C.) At the end of the 3-hour heating period, the paper shall show no tendency to stick to the backing sheet, no evidence of brittleness, and when compared with the original unheated portions of the sample, there shall be no appreciable change of color or appreciable change in the backing sheet.

4.3.5 *Duplicating quality of accelerated-aged stencil paper.*—Take one sheet of stencil paper from each package, and cut it down the middle making two pieces approximately 4½ inches wide and 12 inches long over-all. Save one-half of each sheet as a control. Take the other half, remove the backing sheet and place the stencil sheet in an oven in which the temperature is maintained uniform throughout and heat at 105° ± 2° C. for 1 hour. Then mount the two complementary halves of each stencil sheet on one backing sheet and cut the stencil as described in 4.3.8. on each half. Place the two halves as a unit on a Model #100 mimeograph machine and run at a speed of 5,000 impressions per hour until approximately 1,000 copies are prepared. There shall be no difference of duplicating properties of the heated and unheated portions.

4.3.6 *Cutouts.*—Use an Electromatic typewriter with elite type, previously cleaned, and having a soft platen roll, I.B.M. designation Q.I. The power roll shall have a speed of 280 revolutions per minute, measured with a tachometer, which corresponds approximately to a setting of 8 on the speed indicator. Allow the typewriter to run not less than 15 minutes, then cut the stencil paper with the entire alphabet of both capital and small letters, repeating each letter not less than 25 times. There shall be no evidence of cut-outs either before or after a test run is made on a duplicating machine. After the stencil is cut, clean the platen roll by wrapping a sheet of blotting paper around it and rolling it back and forth several times. Also release the feed rolls.

4.3.7 *Type filling.*—Use an L. C. Smith or a Royal typewriter, elite type, and run off a block of small e's (approximately 300). The last impression shall be as clear and distinct as the tenth impression.

4.3.8 *Correction properties, tracing ability, and printing quality.*—A representative stencil shall be cut on an L. C. Smith typewriter, elite type, utilizing approximately one-fourth of the stencil. The same matter shall then be repeated using a Royal typewriter and an Electromatic typewriter. The latter shall conform to the requirements in 4.3.6 and shall be used as specified therein. Type with the Royal the first line only of the above matter. Take correction compound complying with Federal Specification TT–C–621 and block out the typed line. Now write over the corrected line using an entirely new sentence. In the lower unused portion of the stencil paper trace various circular and rectangular figures using a minimum looped stylus. Place the cut stencil in a Model #100 mimeograph machine and run off 5,000 copies in one hour, using the standard Government Printing Office ink. Retain the first 50 copies, the 100th and 500th, and each additional 500th copy up to and including the 5000th copy. Examine retained copies for correction properties, tracing ability, and printing quality. Failure of the stencil paper to take cor-

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rections properly, failure to work satisfactorily with a stylus, or failure to show satisfactory duplicating quality will be considered as noncompliance with the requirements of this specification.

4.3.9 *Legibility of typing.*—Use an L. C. Smith or a Royal elite-type typewriter and type the sentence, "The quick sly fox jumped over the lazy brown dog's back", using lower case letters. Space up one space and repeat the same sentence using upper case or capital letters. Examine the typing for ease of reading with natural light alone and with artificial plus natural light. Note particularly if any glare is given off by stencil sheet background. If the stencil sheet gives off more than a negligible amount of glare and is difficult to read, the stencil paper shall be considered as failing to meet the requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. — Unless otherwise specified, stencil sheets shall be packaged one quire (24 sheets) to a folded box, wrapper, or envelope with not less than 8 coated or impregnated cushion sheets or 12 noncoated or nonimpregnated cushion sheets. Each package shall contain detailed instructions as specified in 3.16.

5.2 Packing.—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Issue packages.* — Issue packages shall be marked with the name of the commodity, the quantity, the size, the name of the manufacturer, the trade name or trademark of the manufacturer, and the date of manufacture.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked on a side or end with the name of the commodity, the size and quantity contained therein as defined by the contract or order under which shipment is made, the name of the contractor (and the name of the manufacturer, if different), and the number of the contract or order.

6. NOTES

6.1 Ordering data. — Purchasing officers should exercise any desired options offered herein, and should specify the size and any special packaging, packing, or marking desired. (See 1.3, 5.1, 5.3, and 7.3.4.)

6.2 Bid samples.—If samples are required with bids, this requirement should be stated in the inquiry or invitation for bids, giving information as to size or quantity of sample.

6.3 Federal specifications do not include all the types, classes, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Patent notice.—When Government drawings, specifications, or other data, are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and other publications, of the issues in effect on the date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

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7.2 **Army and Navy.**

7.2.1 *Applicable specifications and other publications.—*

Federal Specifications:

NN–B–601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.
LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

National Military Establishment Specifications:

JAN–P–106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.
JAN–P–108—Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN–P–125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.
JAN–P–140—Packaging and Packing for Overseas Shipment — Adhesives, Water-Resistant, Case-Liner.

U. S. Army Specification:

100–2—Standard Specification for Making Shipment by Contractors.[1]

Navy Department Specification:

General Specifications for Inspection of Material.[2]

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.[2]

(Copies of specifications required for Army purchases, may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.

(Copies of Federal and National Military Establishment specifications required for Navy purchases, Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C. except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Virginia. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.)

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

7.2.2 *Packing.*

7.2.2.1 *For domestic shipment.* — Twenty quires, packaged as specified in 5.1, shall be packed in a corrugated or solid fiberboard full-overlap carton opened on the small faces, constructed in accordance with Federal Specification LLL–B–631 or LLL–B–636, respectively. Protective pads of 3/20-inch plywood conforming to the material requirements of Federal Specification NN–B–601 shall be placed inside the carton on both sides of the contents, against the two large faces of the carton. All flaps of the fiberboard box shall be sealed with a good quality adhesive applied evenly throughout the entire area of contact between the flaps.

7.2.2.2 *For overseas shipment.* — Two boxes, each containing 20 quires, packed as specified in 7.2.2.1, shall be overpacked in a full-overlap box constructed, sealed, and strapped in accordance with NME Specification JAN–P–108, compliance symbol V3c or V3s for type 2 load. Arrangement of the inner boxes shall be:

2 in length 1 in width 1 in height.

7.2.2.2.1 Alternatively, two boxes, each containing 20 quires, packed as specified in 7.2.2.1, shall be overpacked in boxes constructed, sealed, and strapped in accordance with NME Specification JAN–P–106, style 4, for type 2 load. Each nailed wood box shall be provided with a sealed case liner or prefabricated bag made of waterproof barrier material conforming to NME Specification JAN–P–125, type L–2 or M. Closure and seams shall be effected by the use of a water-resistant adhesive conforming to NME Specification JAN–P–140, in such a manner as to provide water resistance equal to that of the body material.

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7.2.3 *Marking.*—In addition to the marking specified in 5.3, shipping containers for the Army shall be marked to conform to the requirements of U. S. Army Specification 100–2, except that corner service color marking need not be applied for domestic shipments; for the Navy, in accordance with the Navy Shipment Marking Handbook.

NAVY: *MCS*

7.3 **Air Force.**—Exterior shipping containers for the U. S. Air Force shall be marked in accordance with U. S. Army Specification 94–40645, Marking, Exterior, Domestic and Export Shipment, by Contractors.

(Copies of specifications required for Air Force purchases may be obtained from the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio.)

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

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For sale by the Superintendent of Documents, Washington 25, D. C. Price 5 cents.

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